Tabular Foundation Models: The Next Revolution in Data Science

RAMH workshop

David Holzmüller (SODA Team, INRIA Saclay)

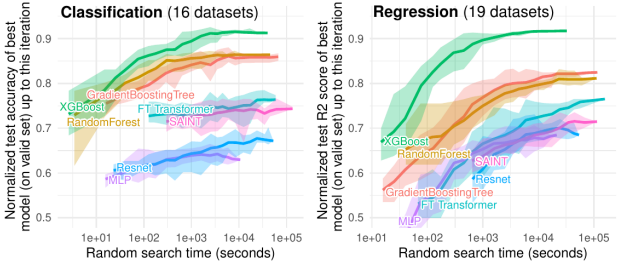
January 27, 2026

Tabular data

Age	Gender	Weight	Time in Hospital	...	Readmitted?
60	Male	90	5 days	...	Yes
45	Female	N/A	3 days	...	No
⋮	⋮	⋮	⋮		⋮

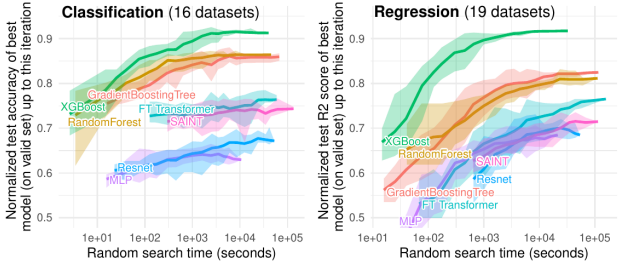
Tabular-data is wide-spread in health & other fields
Much focus on handling numericals & categoricals
Text, dates etc. also common, can be encoded with `skrub`

Tabular data: not easy for deep learning



Source: Grinsztajn et al. [2022]

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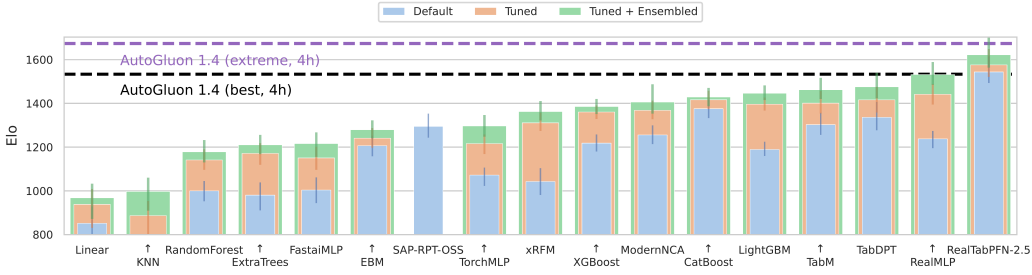


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Typical properties of tabular data:

- no symmetries
- unimportant features
- axis-alignment (random rotations remove useful info)
- noise
- non-smoothness?

Current state of the art on TabArena [Erickson et al., 2025]



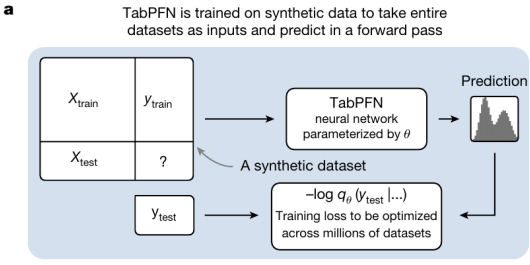
Source: `tabarena.ai`

What are tabular foundation models?

Foundation model = typically large deep learning model, pre-trained on large amounts of data, broadly applicable

- What can we learn from other data?
  - directly related tables (e.g., same database)
  - Semantic understanding (text)
  - **How to best learn from data (meta-learning)**
- Can we just use LLMs? only for small datasets

Tabular foundation models: short overview



Source: Hollmann et al. [2025]

- The NN represents the entire learning algorithm!
- Through pre-training, the NN learns how to learn from new data

Tabular foundation models are easy to use

Installation:

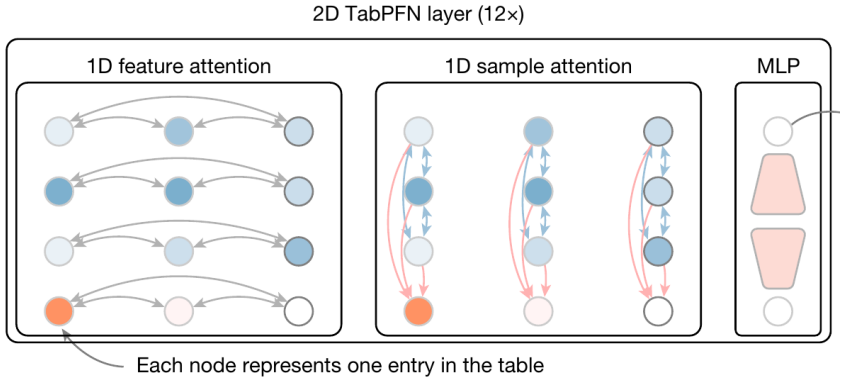
```
pip install tabicl
```

Usage:

```
from tabicl import TabICLClassifier

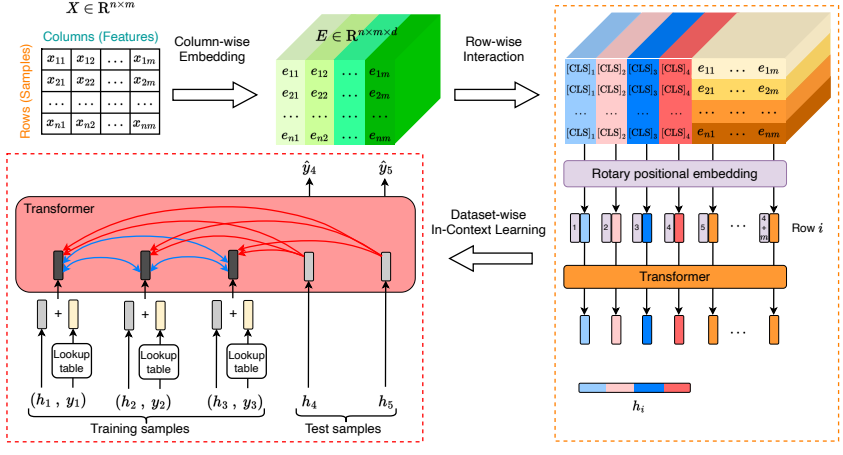
clf = TabICLClassifier()  # downloads checkpoint (108 MB)
clf.fit(X, y)  # only stores the data!
y_pred = clf.predict_proba(X_test)  # calls model(X, y, X_test)
```

TabPFNv2 architecture



TabPFNv2 architecture [Hollmann et al., 2025]

TabICLv1 architecture



TabICLv1 architecture [Qu et al., 2025]

Tabular foundation models: history

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- LimiX, Mitra, Orion-MSP, ConTextTab, etc.

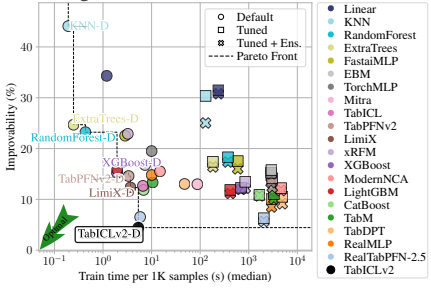
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- LimiX, Mitra, Orion-MSP, ConTextTab, etc.
- (Real)TabPFN-2.5: better, up to 100K samples, non-commercial, nothing known about pre-training+data

TabICLv2: sneak peek

Joint work with: Jingang Qu, Gaël Varoquaux, Marine Le Morvan

TabICLv2: Preliminary results

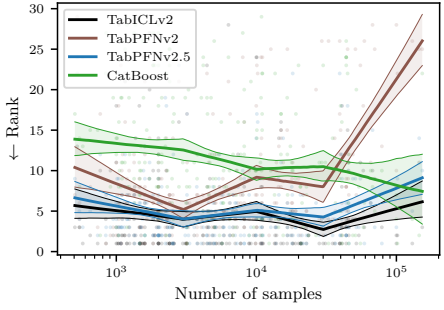


Results on TabArena [Erickson et al., 2025].

TabICLv2 matches RealTabPFN-2.5

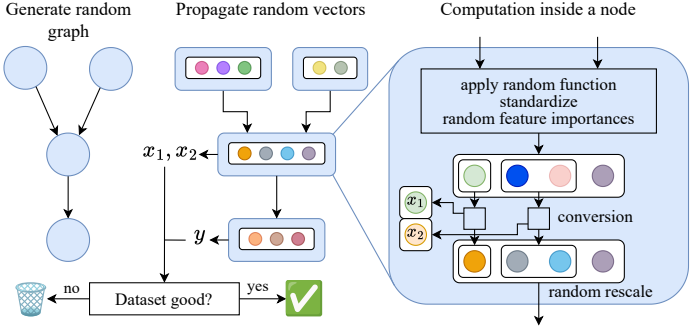
- even without tuning+ensembling
- even without real-data post-training
- with moderate pre-training compute ($\approx$ 25 H100-days)
- while being fully open (model soon, pre-training after acceptance)

TabICLv2 scales to large datasets

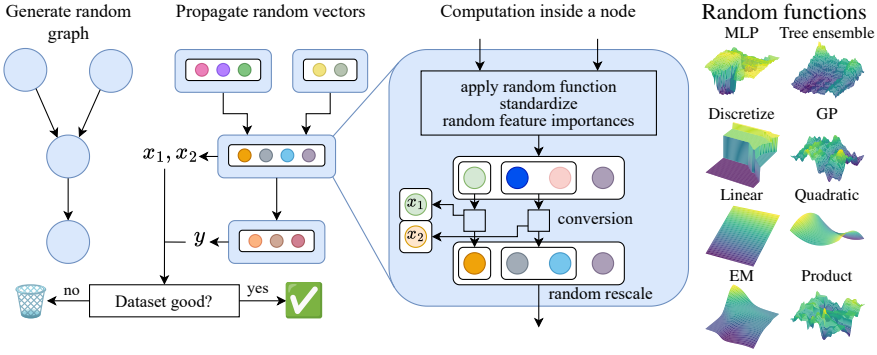


Rankings on TALENT [Ye et al., 2024] depending on the number of samples.

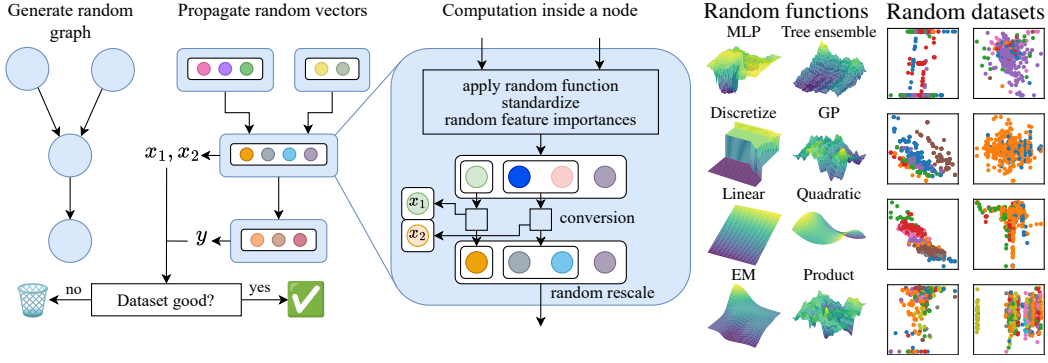
TabICLv2 synthetic data generation (prior)



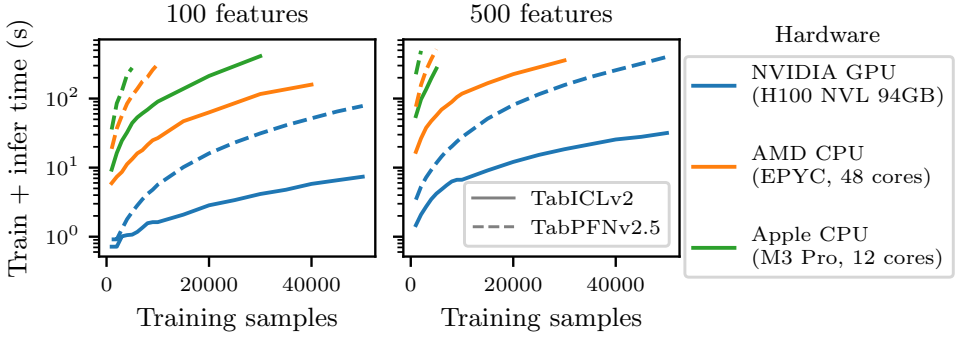
TabICLv2 synthetic data generation (prior)



TabICLv2 synthetic data generation (prior)



TabICLv2 runtime



Fast for:

- GPU
- Smaller datasets
- Training

Slow for:

- CPU
- Larger datasets
- Inference

Squeezing out maximum accuracy

- Even tabular foundation models have hyperparameters: preprocessing, temperature, checkpoint, ...
- Tuning + ensembling often helps a bit
- Fine-tuning on one or multiple datasets
- Tuning + ensembling over multiple models $\rightarrow$ AutoML tools like AutoGluon

Related trends

- Multi-table foundation models

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- Causal inference (& discovery??) $\rightarrow$ Treatment effect estimation
- Survival analysis??
- Probabilistic inference? Black-box optimization? Epistemic uncertainties?
- Simulation-based inference

Conclusion

- Tabular foundation models are now state-of-the-art up to $\approx$ 100K train samples
- Trade-offs: Slow on CPU + large datasets
- Many promising directions for extensions

Literature I

Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? *Neural Information Processing Systems*, 2022.

Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, David Salinas, and Frank Hutter. TabArena: A living benchmark for machine learning on tabular data. In *Neural Information Processing Systems*, 2025.

Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. URL https://www.nature.com/articles/s41586-024-08328-6. Publisher: Nature Publishing Group UK London.

Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations*, 2023.

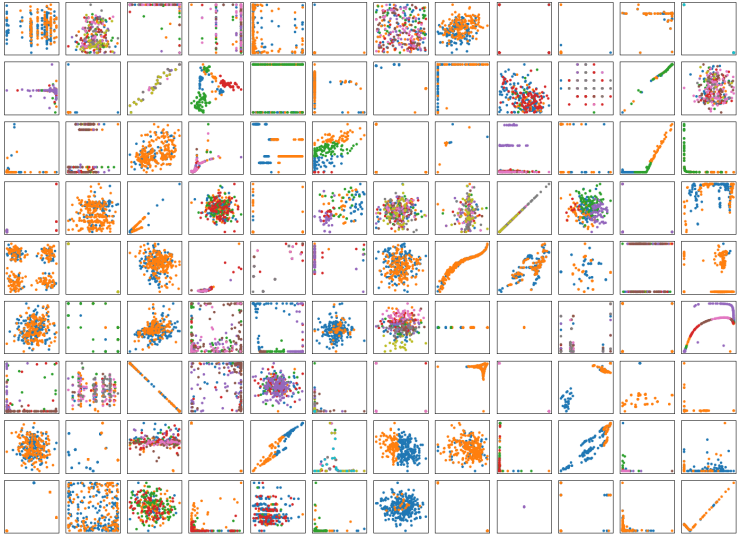
Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICL: A Tabular Foundation Model for In-Context Learning on Large Data, February 2025. URL http://arxiv.org/abs/2502.05564. arXiv:2502.05564 [cs].

Literature II

Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Benjamin Jäger, Dominik Safaric, Simone Alessi, Adrian Hayler, Mihir Manium, Rosen Yu, Felix Jablonski, Shi Bin Hoo, Anurag Garg, Jake Robertson, Magnus Bühler, Vladyslav Moroshan, Lennart Purucker, Clara Cornu, Lilly Charlotte Wehrhahn, Alessandro Bonetto, Bernhard Schölkopf, Sauraj Gambhir, Noah Hollmann, and Frank Hutter. TabPFN-2.5: Advancing the State of the Art in Tabular Foundation Models, November 2025. URL http://arxiv.org/abs/2511.08667. arXiv:2511.08667 [cs].

Han-Jia Ye, Si-Yang Liu, Hao-Run Cai, Qi-Le Zhou, and De-Chuan Zhan. A closer look at deep learning on tabular data. *arXiv:2407.00956*, 2024.

TabICLv2 synthetic datasets (unfiltered)



Tabular foundation models: general trade-off

+ Very high accuracy, even without tuning, on small–medium datasets (≤ 50K samples)
+ Hard to use it wrong (no early stopping / tricky hyperparameters)
+ Fast on smaller datasets *on GPUs*
- Slow on CPUs
- Slow inference (KV caching or distillation can help)
- Large datasets: slow, high memory usage, hard to pre-train well
- Some models have restrictive licenses
- Black-box (most of them)